

KAVIRATHNA VELMURUGAN

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PROFESSIONAL SUMMARY

Final-year AI & Data Science student (CGPA 8.01) who ships production-grade AWS-serverless and RAG/LLM applications end-to-end. Cut retrieval latency ~40% on a FAISS-based semantic search pipeline and earned an Outstanding (Grade O) rating in AWS Educate's ML-integrated cloud engineering program.

PROJECTS

- Department Notice Portal** notice-portal-six.vercel.app
React, TypeScript, AWS Cognito, Lambda, DynamoDB, S3, SES, Groq
 - Replaced ad-hoc, department-wide notice circulation with a role-based portal serving 3 user tiers (HOD, faculty, students), adding AI-assisted drafting and scheduled publishing via Groq.
 - Architected a 5-service serverless AWS backend (Cognito, Lambda, DynamoDB, S3, SES) and deployed it live in the ap-south-1 (Mumbai) region.
- FarmCredit AI** farmcredit-m3o9ke6y7zquszdbitqhe4.streamlit.app
FastAPI, XGBoost, SHAP, QLoRA, Mistral, Streamlit
 - Built a crop-loan risk prediction model (XGBoost) with SHAP-based explainability, giving lenders transparent, feature-level reasoning behind each risk score.
 - Fine-tuned a QLoRA advisory assistant on Mistral using a 500-example synthetic dataset spanning 5 domains (risk explanation, crop suggestions, scheme eligibility, repayment plans, farmer Q&A), deployed live via Streamlit.
- BotStudio** bot-studiov1.vercel.app
Next.js, Supabase, pgvector, OpenRouter
 - Engineered a no-code RAG chatbot builder with pgvector-based semantic retrieval, enabling non-technical users to stand up a context-aware bot without writing code.
 - Shipped an embeddable chat widget for third-party sites, packaging the retrieval pipeline behind a single drop-in script.
- DataPilot AI** datapilot-ej9z.vercel.app
React, FastAPI, Pandas, Groq, MongoDB
 - Built a spreadsheet-analysis SaaS that automates data cleaning on upload, removing the manual pre-processing step users previously did by hand.
 - Generated AI-powered dataset insights via Groq, surfacing patterns from uploaded data without any manual query-writing.

EXPERIENCE

- Full Stack MERN Developer Intern — Skill Hive Innovations, Coimbatore** May 2026 – Present
 - Leading full-stack development of Invita, Skill Hive Innovations' first live client engagement — a community and family management platform (React, Node.js, MongoDB, Flutter).
 - Engineered reusable Next.js components and REST APIs after a competitive analysis of global and Indian community-management platforms, speeding up delivery across client projects.
- AI/ML Developer Intern — Techpuram, Madurai (Remote)** Nov 2025 – May 2026
 - Engineered AST-parsing and FAISS vector-embedding pipelines that improved retrieval speed by ~40% over the prior approach.
 - Built Code Descriptor, a GitHub repo explainer chaining MiniLM embeddings, FAISS retrieval, and FLAN-T5/DeepSeek Coder generation to auto-summarize unfamiliar codebases.
- AI Cloud Engineer Virtual Intern — AWS Educate** Jan 2026 – Mar 2026
 - Completed the program with Grade O (Outstanding), its top rating, for ML-integrated cloud solution design.
 - Designed and deployed ML-integrated solutions across core AWS services (compute, storage, and inference) in the program's applied labs.

EDUCATION

- B.Tech, Artificial Intelligence & Data Science — SSM Institute of Engineering and Technology, Dindigul** Expected May 2027
CGPA: 8.01

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, HTML, CSS
AI/ML: PyTorch, XGBoost, SHAP, Scikit-learn, FAISS, RAG, LangChain, Groq, QLoRA, Mistral
Web: React, Next.js, Node.js, FastAPI, Express, Tailwind CSS, Socket.io, Vite
Cloud & DB: AWS (Cognito, Lambda, API Gateway, DynamoDB, S3, SES), MongoDB, PostgreSQL, Supabase, Vercel
Tools: Git, GitHub, Docker, Postman

CERTIFICATIONS

- AWS Certified AI Practitioner — Amazon Web Services** Nov 2025 – Nov 2028
- AWS Certified Cloud Practitioner — Amazon Web Services** Mar 2026 – Mar 2029